PAPER ID-310980

Subject Code: BCE070

Roll No: | | | | | | | | | | | | | |

BTECH
(SEM VII) THEORY EXAMINATION 2025-26
RAILWAY,AIRWAY AND WATERWAY INFRASTRUCTURE SYSTEM

TIME: 3 HRS **M.MARKS: 70**

Note: Attempt all Sections. In case of any missing data; choose suitably.

SECTION A

1. Attempt *all* questions in brief. **02 x 7 = 14**

Q no.	Question	CO	Level
a.	Define railway gauge.	1	K1
b.	What are points and crossings?	1	K1
c.	Define interlocking.	2	K1
d.	What is the purpose of automatic signals?	2	K1
e.	What is a holding apron?	3	K1
f.	Define groyne.	4	K1
g.	Define inland navigation.	5	K1

SECTION B

2. Attempt any *three* of the following: **07 x 3 = 21**

a.	Describe the essential elements of railway alignment. What factors influence the selection of alignment?	1	K2
b.	Discuss various types of level crossings and their safety measures.	2	K2
c.	Explain the concepts of airport capacity and the factors influencing it.	3	K2
d.	Describe the basic principles and components involved in the ship berthing process.	4	K2
e.	Explain rehabilitation techniques used for aging or damaged transport infrastructure.	5	K2

SECTION C

3. Attempt any *one* part of the following: **07 x 1 = 07**

a.	If the wheel base of a vehicle moving on a B.G. track is 6m, the diameter of wheel is 1.5 m and the depth of flanges below the top of rail is 3.17 cm. Determine the extra width required to be provided on gauge, if the radius of the curve is 160m.	1	K3
b.	Discuss various track maintenance methods such as manual maintenance, mechanical tamping, rail grinding, and lubrication. Explain their advantages and limitations.	1	K2

4. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Describe in detail the types of railway signals (semaphore, colour light, automatic, and cab signals). Explain their functions with neat sketches.	2	K2
b.	Explain disaster prevention, preparedness, and mitigation strategies adopted by Indian Railways. Describe the role of emergency response teams.	2	K2

5. Attempt any *one* part of the following: **07 x 1 = 07**

a.	Write a detailed note on airport layout planning and the interrelationship between runway systems, taxiways, aprons, and terminals.	3	K2

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b.	For the hottest month of the year at the proposed airport site, the mean of the average daily temperature is 47^0C. Calculate the airport reference temperature. If the site is at mean sea level with a ground level, calculate the actual runway length to be provided.	3	K3

6. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain natural and artificial harbours in detail. Compare their features, advantages, disadvantages, and suitability with real-world examples.	4	K2
b.	What is a harbour? Compare natural harbours and artificial harbours with examples.	4	K2

7. Attempt any *one* part of the following: 07 x 1 = 07

a.	Explain channel alignment in detail. Discuss the factors influencing alignment such as topography, hydrology, soil characteristics, and environmental considerations.	5	K2
b.	Write a detailed note on the combined role of channel alignment, dredging, and maintenance in ensuring efficient inland water transport.	5	K2

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